In-Class Activity: Multiple IV Infusions

Module 6 - Multiple Dosings · In-class activity · 28 Sep 2026

Name _____ Date _____

1a A patient is to receive multiple intravenous (IV) infusions of 500 mg of a drug administered over a period of two hours. The second 2-hour infusion starts six hours after the start of the previous infusion. The reported elimination half-life of the drug is 3 hours, and the apparent volume of distribution is 18 L. What is the expected plasma drug concentration at the end of the first infusion?

1b A patient receives 500 mg of a drug by intravenous (IV) infusion over two hours, and a second 2-hour infusion of 500 mg starting six hours after the start of the first ($t_{½}$ 3 hours, V_D 18 L). Each infusion reaches 22.24 mg/L at its end. What is the concentration of drug in the plasma four hours after the end of the second infusion?

2a 150 mg of a drug (Cl = 2.54 L/hr, V_D = 22 L) was infused intravenously over 1.5 hours. Exactly eight hours after the start of the first infusion, a second 150-mg dose was again infused over 1.5 hours. What is the plasma drug concentration at the end of the first infusion?

2b 150 mg of a drug (Cl = 2.54 L/hr, V_D = 22 L) was infused intravenously over 1.5 hours, and a second 150-mg dose was infused over 1.5 hours starting exactly eight hours after the start of the first. Each infusion reaches 6.26 mg/L at its end. What is the concentration of drug in the plasma six hours after the end of the second infusion?

2c A drug with a clearance of 2.54 L/hr is being given as 150-mg intravenous (IV) infusions over 1.5 hours. If the drug were continuously infused at the rate above, what would the steady-state concentration be?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a Her key

Answer: 22.24 mg/L

WHAT THE STEM GIVES

D_0	500 mg	with the infusion time gives R = $\frac{500}{2}$ = 250 mg/hr
infusion time	over a period of two hours	R = 250 mg/hr; also t = 2 hr in e^{-kt}
second start	six hours after the start of the previous infusion	*not needed: the first infusion ends before it begins*
t½	3 hours	gives k = $\frac{0.693}{3}$ = 0.231 hr^{-1}
V_D	18 L	in Cl = kV_D = 0.231 × 18 = 4.158 L/hr

THE WORKING, ONE LINE AT A TIME

UNIT

$$R = \frac{500\ mg}{2\ hr} = 250\ mg/hr;\ k = \frac{0.693}{3\ hr} = 0.231\ hr^{-1}$$

The equation takes a rate and a rate constant. The dose over its duration gives the rate in mg/hr, and the half-life gives k in reciprocal hours.

SETUP

$$C_p = \frac{R}{V_D\ k}\ (1 - e^{-kt}),\ \text{with } t = 2\ hr$$

This is the rise during one infusion, stopped at the infusion time. The 6-hour spacing of the infusions plays no part in this first value.

ALGEBRA

$$V_D\ k = (18\ L)(0.231\ hr^{-1}) = 4.158\ L/hr;\ \frac{R}{V_D\ k} = \frac{250\ mg/hr}{4.158\ L/hr} = 60.13\ mg/L$$

60.13 mg/L is the plateau a continuous infusion at 250 mg/hr would reach after 3 to 5 half-lives. mg/hr over L/hr leaves mg/L; the 2-hour infusion reaches only a fraction of it.

ALGEBRA

$1 - e^{-(0.231)(2)} = 1 - e^{-0.462} = 1 - 0.630 = 0.370$; $C_p = 60.13 \times 0.370 = 22.25$ mg/L

Two hours is two thirds of the 3-hour half-life, so 37 per cent of the plateau is reached when the infusion stops. The exponent, 0.231 × 2, is a pure number.

ROUND

$$C_p = \frac{250 \text{ mg/hr}}{4.158 \text{ L/hr}}(1 - e^{-0.462}) = 22.24 \text{ mg/L}$$

Rounded to 60.13 and 0.370 the factors give 22.25; carried unrounded they give 22.2449, which rounds to 22.24 mg/L, her confirmed "something like 22.2 mg per liter for the first part for A". Because the second infusion is identical, it is the C_0 for both declines in part (b).

> **DOES THE NUMBER MAKE SENSE?**
>
> The plateau R over Cl is $\frac{250}{4.158} = 60.13$ mg/L. Two hours is two-thirds of a 3-hour half-life, so $1 - 0.630 = 0.370$ of the plateau is reached: 22.24 mg/L, well below 60.13.

1b Her key

Answer: 11.04 mg/L

WHAT THE STEM GIVES

D_0	500 mg	*not needed: the end-of-infusion level is given*
infusion time	over two hours	fixes when each infusion ends: 2 hr and 8 hr
second start	six hours after the start of the first	the second infusion runs 6 to 8 hr
t½	3 hours	gives $k = \frac{0.693}{3} = 0.231$ hr^{-1}
V_D	18 L	*not needed: the end-of-infusion level is given*
Cend	22.24 mg/L	the starting level of each decay term
t	four hours after the end of the second infusion	the second term decays 4 hr, the first 10 hr

THE WORKING, ONE LINE AT A TIME

SETUP

Time line: first infusion 0 to 2 hr; second 6 to 8 hr; time asked for 8 + 4 = 12 hr

The times match her Example 4: the infusions end at 2 and 8 hours, and the question asks about 12 hours. Each infusion then declines from its own end.

SETUP

$C = 22.24\ e^{-0.231(12\ -\ 8)} + 22.24\ e^{-0.231(12\ -\ 2)}$

The second infusion has declined for 4 hours and the first for 10. Both start from 22.24 mg/L, because both infusions were the same dose over the same time.

ALGEBRA

Second: $22.24 \times e^{-0.924} = 22.24 \times 0.397 = 8.83$ mg/L

kt = 0.231 × 4 = 0.924, a pure number. Four hours is 1.33 half-lives of 3 hours, so about 40 per cent of the second infusion remains at the time asked for.

ALGEBRA

First: $22.24 \times e^{-2.31} = 22.24 \times 0.0993 = 2.21$ mg/L

kt = 0.231 × 10 = 2.31. Ten hours is 3.33 half-lives, so about a tenth of the first infusion remains; it ended 6 hours before the second did.

ALGEBRA

C = 8.83 + 2.21 = 11.04 mg/L

The sum of the two contributions. Most of it comes from the second infusion, because the first has had 6 more hours to be eliminated.

ROUND

C = 11.04 mg/L

Her confirmed value was "eleven-ish", with rounding differences between students. Carried from 22.24, the sum is 11.04 mg/L.

DOES THE NUMBER MAKE SENSE?

Four hours is a little over one 3-hour half-life, so the second infusion alone leaves a bit under half of 22.24 mg/L, 8.83; the first, 10 hours on, adds 2.21 more. The sum lies between 8.83 and 2 × 22.24 mg/L.

2a

Her key

Answer: 6.26 mg/L

WHAT THE STEM GIVES

D_0	150 mg	with the infusion time gives R = $\frac{150}{1.5}$ = 100 mg/hr
Cl	2.54 L/hr	the denominator of R over Cl, and the

		numerator of $k = \frac{Cl}{V_D}$
V_D	22 L	gives $k = \frac{2.54}{22} = 0.1155\ hr^{-1}$ for the exponent
infusion time	over 1.5 hours	R = 100 mg/hr; also t = 1.5 hr in e^{-kt}
second start	Exactly eight hours after the start of the first infusion	*not needed: the first infusion is over before it begins*

THE WORKING, ONE LINE AT A TIME

UNIT

$$R = \frac{150\ mg}{1.5\ hr} = 100\ mg/hr;\ k = \frac{Cl}{V_D} = \frac{2.54\ L/hr}{22\ L} = 0.1155\ hr^{-1}$$

This stem gives the clearance (Cl), not k or a half-life, so k comes from $Cl = kV_D$. Litres cancel, leaving reciprocal hours.

SETUP

$$C_p = \frac{R}{Cl}(1 - e^{-kt}),\ \text{with } t = 1.5\ hr$$

With the clearance given, $\frac{R}{Cl}$ is the direct form of the plateau; it equals $\frac{R}{V_D\,k}$. t is the infusion time.

ALGEBRA

$$\frac{R}{Cl} = \frac{100\ mg/hr}{2.54\ L/hr} = 39.37\ mg/L;\ 1 - e^{-(0.1155)(1.5)} = 1 - e^{-0.173} = 1 - 0.841 = 0.159$$

kt = 0.1155 × 1.5 = 0.173, a pure number. An hour and a half is a quarter of the 6-hour half-life ($\frac{0.693}{0.1155}$), so only 16 per cent of the plateau is reached.

ALGEBRA

$$C_p = 39.37 \times 0.159 = 6.26\ mg/L$$

This is 16 per cent of 39.37 mg/L. The plateau is high because the clearance is small, and the infusion is short against the half-life, so the product is modest.

ROUND

$$C_p = 6.26\ mg/L$$

Her value, 6.26 mg/L. It is the C_0 for the decline after this infusion and, because the second infusion is identical, for the second decline as well.

DOES THE NUMBER MAKE SENSE?

The plateau R over Cl is $\frac{100}{2.54}$ = 39.37 mg/L; 1.5 hours is a quarter of the 6-hour half-life, so only 0.159 of the plateau is reached: 6.26 mg/L, far below 39.37.

2b

Worked by the drill

Answer: 4.37 mg/L

WHAT THE STEM GIVES

D_0	150 mg	*not needed: the end-of-infusion level is given*
Cl	2.54 L/hr	with V_D gives k = $\frac{2.54}{22}$ = 0.1155 hr^{-1}
V_D	22 L	the denominator of k
infusion time	over 1.5 hours	fixes when each infusion ends: 1.5 hr and 9.5 hr
second start	exactly eight hours after the start of the first	the second infusion runs 8 to 9.5 hr
Cend	6.26 mg/L	the starting level of each decay term
t	six hours after the end of the second infusion	the second term decays 6 hr, the first 14 hr

THE WORKING, ONE LINE AT A TIME

SETUP

Time line: first infusion 0 to 1.5 hr; second 8 to 9.5 hr; time asked for 9.5 + 6 = 15.5 hr

The second infusion starts at 8 hours and runs 1.5 hours, so it ends at 9.5; six hours later is 15.5 hours. She noted the temptation to call it 16.

SETUP

$C = 6.26\ e^{-k(15.5\ -\ 9.5)} + 6.26\ e^{-k(15.5\ -\ 1.5)}$, with $k = 0.1155\ hr^{-1}$

The second infusion has declined for the 6 hours given; the first for 15.5 − 1.5 = 14 hours, back to when it ended.

ALGEBRA

Second: $6.26 \times e^{-0.693} = 6.26 \times 0.500 = 3.13$ mg/L

kt = 0.1155 × 6 = 0.693, exactly one half-life, so half of the second infusion remains six hours after it ends. $e^{-0.693}$ is 0.500.

ALGEBRA

First: $6.26 \times e^{-1.617} = 6.26 \times 0.1985 = 1.24$ mg/L

kt = 0.1155 × 14 = 1.617, which is 2.33 half-lives, so about a fifth of the first infusion remains. The 14 is 15.5 − 1.5, measured from the end of the first infusion.

ALGEBRA

C = 3.13 + 1.24 = 4.37 mg/L

The sum of the two contributions. The second infusion supplies 3.13 of the 4.37 mg/L, because it ended 8 hours later than the first.

ROUND

C = 4.37 mg/L

Computed from her inputs. With k rounded to 0.115, the pieces are 3.14 and 1.25 and the sum is 4.39; both sums lie inside the tolerance.

> **DOES THE NUMBER MAKE SENSE?**
> k = 0.1155 hr^{-1} is a 6-hour half-life, so six hours after the second infusion exactly half of 6.26 mg/L, 3.13, remains; the first infusion, 14 hours on, adds 1.24. The sum lies between 3.13 and 2 × 6.26 mg/L.

2c Her key

Answer: 39.37 mg/L

WHAT THE STEM GIVES

Cl	2.54 L/hr	the denominator
D_0	150-mg	with the infusion time gives R = $\frac{150}{1.5}$ = 100 mg/hr
infusion time	over 1.5 hours	the denominator of R

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R}{Cl}$$

A continuous infusion reaches a plateau where the rate in equals the rate out, and the rate out is clearance times concentration. The interval does not enter, because the infusion never stops.

UNIT

$$R = \frac{150 \text{ mg}}{1.5 \text{ hr}} = 100 \text{ mg/hr}$$

The same rate the short infusions run at, 150 mg over 1.5 hours, now imagined running without a break. mg per hr is the zero-order input the plateau depends on.

ALGEBRA

$$C_{ss} = \frac{100 \text{ mg/hr}}{2.54 \text{ L/hr}} = 39.37 \text{ mg/L}$$

mg/hr over L/hr leaves mg/L. The 1.5-hour infusions reach only 16 per cent of this $\frac{R}{Cl}$, because they stop long before 3 to 5 half-lives have passed.

ROUND

C_{ss} = 39.37 mg/L

Her method, the rate divided by the clearance, gives 39.37 mg/L. It has no time term, so it says nothing about when the plateau would be reached.

DOES THE NUMBER MAKE SENSE?

Each hour 100 mg enters and 2.54 L of plasma is cleared; at the plateau they balance, so $C_{ss} = \frac{100 \text{ mg/hr}}{2.54 \text{ L/hr}} = 39.37$ mg/L. The rate is above zero, so C_{ss} is too.